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An SAO-based text mining approach to building a technology tree for technology planning

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ABSTRACT

A technology tree (TechTree) is a branching diagram that expresses relationships among product components, technologies, or functions of a technology in a specific technology area. A TechTree identifies strategic core technologies and is a useful tool to support decision making in a given market environment for organizations with specified capabilities. However, existing TechTrees generally overemphasize qualitative and expert-dependent knowledge rather than incorporating quantitative and objective information. In addition, the traditional process of developing a TechTree requires vast amounts of information, which costs considerably in terms of time, and cannot provide integrated information from a variety of technological perspectives simultaneously. To remedy these problems, this research presents a text mining approach based on Subject–Action–Object (SAO) structures; this approach develops a TechTree by extracting and analyzing SAO structures from patent documents. The extracted SAO structures are categorized by similarities, and are identified by the type of technological implications. To demonstrate the feasibility of the proposed approach, we developed a TechTree regarding Proton Exchange Fuel Cell technology.

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1. Introduction

A technology tree (TechTree) is a branching diagram (tree structure) that represents relationships among technologies. Generally, a TechTree is used to represent the relationships among product components, technologies, or functions of a technology in a specific technology area. A TechTree is a useful tool for supporting a decision making process to identify strategic core technologies in a given market environment and organization capabilities. Furthermore, it helps to define element technologies based on identified core technologies. Due to these advantages, TechTrees are important decision-making tools for technology planning (Cheong, 2006), and are utilized in technology planning (Aude & Kahn, 1986; Choudhury & Fallah, 2009; Durand, 1992; Guglielmi, Williams, Groepper, & Lascar, 2010; Visentin, 2008; Yoon, Phaal, & Probert, 2008).

The current process of developing a TechTree has the following limitations. The first is exclusive dependence on experts during the development process. Currently, most TechTree diagrams are created by reflecting the opinions of technology domain experts. However, developing a TechTree by this process has considerable costs in terms of time and human resources; and obtaining the participation of an expert can be difficult (Heiss & Jankowsky, 2001). Recently, new technologies have continuously appeared and replaced old technology. To establish strategic technology planning that considers the rapid technological changes, a TechTree should be constantly updated to reflect the changes. However, because of the current process, updating and managing TechTrees are difficult. The second limitation is that current TechTrees only represent limited technology information. A TechTree can represent various technology perspectives such as product taxonomy (Aude & Kahn, 1986; Cascini & Zini, 2008; Durand, 1992; Yoon et al., 2008), technology taxonomy (Choudhury & Fallah, 2009; Durand, 1992; Guglielmi et al., 2010; Visentin, 2008), and function taxonomy of technology (Cascini & Zini, 2008; Cheong, 2006). However, the TechTree represents technology information from only the technology perspective for which the tree was developed. Another limitation is that existing TechTrees cannot represent subsidiary information related to technologies. In technology analysis, the subsidiary information of technology is as important as the technology itself. We called this 'technology meta-information'. This information includes the inventor of the technology,

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the time at which the technology was developed, or where it was developed. By integrating these various perspectives of technology and technology meta-information, we can generate a TechTree that helps decision-making in technology planning more effectively than do existing TechTrees.

To address these problems, we suggest using Subject–Action–Object (SAO)-based text mining techniques to develop TechTrees. This approach analyzes patent documents and exploits the analysis results to create a TechTree that represents technology perspectives. Patent information includes valuable up-to-date technological information and is continuously updated. Patents also include bibliometric information and this information can be utilized as technology meta-information.

The proposed approach consists of a procedure for constructing source data from patents and a method of developing a TechTree using the data. The procedure uses Natural Language Processing (NLP) technology to extract SAO structures from patents, and text-mining techniques to analyze the SAO structures. The resulting information is used as the source data for development of a technology tree. The method develops a TechTree that represents various technology perspectives such as product taxonomy, technology taxonomy, and function taxonomy of technology. The suggested TechTree is represented with technology meta-information.

The rest of paper is organized as follows. In Section 2 we describe work related to the proposed approach. In Section 3 we provide a detailed description of our approach. In Section 4 we illustrate the proposed framework by describing a case study of developing a TechTree that represents the technology used in Proton Exchange Membrane Fuel Cells (PEMFCs). In Section 5 we provide concluding remarks and directions for further study.

2. Related work

2.1. Technology tree

Although a TechTree includes the word "technology", it does not mean the only tree structure for representing technological information. TechTrees can be classified into three categories based on the purpose for which they are developed: product taxonomy, technology taxonomy, and function taxonomy of technology.

A 'product taxonomy' TechTree depicts components used in a product and their interrelatedness. Aude and Kahn (1986) used a TechTree to describe structures of a silicon product, and Yoon et al. (2008) used a TechTree to represent the components of a mobile phone. Other researchers used TechTrees to express product taxonomy information (Buderi, 2000; Cascini & Zini, 2008; Durand, 1992).

A 'technology taxonomy' TechTree depicts technologies applied to a product and their interrelatedness (Lee, Kang, Park, & Park, 2008). Lee et al. (2008) used a TechTree to represent the technology taxonomy of Broadband Network a preliminary step to creating a technology roadmap. The European Space Agency (ESA) used a TechTree to describe technology taxonomy. ESA reviewed the European and worldwide trends in the Space Automation and Robotics field to identify long term development trends, and used a TechTree represent the outcome (Guglielmi et al., 2010; Visentin, 2008). Choudhury and Fallah (2009) used the VND (Veterans, Newbies and Downgraded) model and a "green TechTree" to show the general trajectory followed by the green energy sector.

A 'function taxonomy of technology' TechTree depicts the functions of technology applied to products and technologies and their interrelatedness. The function of technology presents the information regarding the purposes of technologies, the effects of technologies, or the interrelationships of products. Cascini and Zini (2008) used a computer-aided patent analyzer system (PatAnalyzer) to develop functional TechTrees of products. Unlike previous studies, this research examined ways to use computing resources to generate a TechTree. Cheong (2006) suggested a functional TechTree used at the Samsung Advanced Institute of Technology (SAIT). SAIT used the functional TechTree to identify the core functions of technologies by representing the interrelationships among core functions and secondary functions that support the core functions.

2.2. SAO structures for technology analysis

In this paper, an SAO structure is used to construct a TechTree. SAO structures are composed of Subject (noun phrase), Action (verb phrase) and Object (noun phrase). SAO structures are commonly used to represent functions of technology. This simple structure explicitly describes a relationship between components that appear in a patent text. For example, an SAO structure can simply represent the function of a battery as "Battery Energizes Bulb". In this example, "Battery" is the subject, "Energizes" is the action, and "Bulb" is the object. The technology purpose of "Battery" is to energize "Bulb" and the function of "Battery" is "To energize Bulb". Recently, many researchers have attempted to determine how to use SAO structures for technology analysis (Moehrle, 2010; Sternitzke & Bergmann, 2009; Yoon & Kim, 2012; Yoon, Lim, Choi, Kim, & Kim, 2011).

SAO structures can be used to represent a variety of technology information. Subjects and objects may refer to components of a system; actions may refer to functions performed by and on components (Cascini, Fantechi, & Spinicci, 2004a). Moreover, an SAO structure can be organized in a problem–solution format if the action–object (AO) states the problem and the subject (S) forms the solution (Moehrle, Walter, Geritz, & Müller, 2005). SAO structure also states partative relationships among products or technologies (Cascini, Lueehesi, & Rissone, 2001). If the action word is partative verb such as such as "have", "composed of", and "be made of", the component of a subject may include the object as a component.

To computationally extract SAO structure from patent textual descriptions, use of natural language processing (NLP) is necessary (Cascini, Fantechi, & Spinicci, 2004b) in the Part-Of-Speech tagging process. Because previous studies regarding NLP have been already conducted to extract SAO structures from a sentence (Charniak et al., 1996; Choi, Lim, Yoon, & Kim, 2010; Liu & Singh, 2004; Tsourikov, Batchilo, & Sovpel, 2000), we use methods described in those studies to extract SAO structures.

2.3. WordNet based sentence similarity

With SAO structures, we use WordNet-based Sentence Similarity (Simpson & Dao, 2005) to determine extracted SAO structures. To utilize large amounts of SAO structures from patent text, a clustering method is needed that can categorize the SAO structures by using similarities. Because an SAO structure is a simple sentence, the WordNet-based Sentence Similarity method is a useful tool to calculate similarities among SAO structures.

In general, the measurement process of assessing semantic similarity between two sentences is composed of (1) tokenizing the sentences, (2) stemming words, (3) tagging parts of speech, (4) determining the most likely meaning of each word in each sentence, and (5) computing the sentence similarity based on the similarity between pairs of corresponding words (Simpson & Dao, 2005). A measure of similarity (sim) between two concepts (c_1, c_2) is defined as follows (Resnik, 1999):

$$sim(c_1, c_2) = \frac{2 \times depth\,(lcs(c_1, c_2))}{depth\,(c_1) + depth\,(c_2)}, \quad (1)$$

where lcs is the lowest common subsume of c_1 and c_2, and $depth$ is the distance from a concept node c_i to the root of a concept hierarchy, and $0 < sim(c_1, c_2) \leqslant 1$ where 1 means that the concepts are

identical. Building on this measure of similarity, the matching average *MatAvg* (Simpson & Dao, 2005) to compute similarity between sentences X and Y is:

$$MatAvg(X,Y) = \frac{2 \times Match(X,Y)}{|X| + |Y|}, \quad (2)$$

where $|\cdot|$ ($\cdot = X$ or Y) is the number of set tokens in the sentence, $Match(X,Y)$ is the sum of the similarity of the matching word tokens between sentences X and Y, and $0 < MatAvg(X,Y) \leqslant 1$, where 1 means that the sentences are identical. Using the WordNet semantic dictionary (Miller, 1995) as the concept hierarchy, the measure for similarity measurement between two words (1), and the matching average for similarity measurement between two sentences (2), a .NET based semantic sentence similarity measurement has been implemented as a C# library (Simpson & Dao, 2005).

3. An SAO-based text-mining approach to developing a technology tree

In this paper, we suggest an SAO-based text-mining approach for developing a TechTree. The proposed approach consists of a procedure for manipulating SAO structures and a method for constructing a TechTree using processed SAO structures.

3.1. A procedure for manipulating SAO structures

The suggested procedure comprised of six steps: (1) Selecting patent documents, (2) Extracting SAO structures, (3) Measuring SAO similarity, (4) Categorizing word phrases and action-object combinations, (5) Identifying types of word phrases, and (6) Identifying types of action-object combinations (Fig. 1).

3.1.1. Selecting patent documents

The first step is selection of patent documents. The main objective of this step is to select patent documents from a patent database and to eliminate irreverent patents. To collect patents, a patent retrieval query is used. The query consists of keywords related to the target technology, and bibliographic information such as International Patent Code (IPC), applicants and application date. After collecting patents, we eliminate irrelevant patents by applying a filtering rule.

3.1.2. Extracting SAO structures

The second step is extraction of SAO structures from the selected documents. SAO structures can be extracted from any description in textual format including the abstract, claims, and description sections of a patent document. Of these sections, the abstract is precise and has been regarded as the most meaningful part (Chen, Tokuda, & Adachi, 2003). Existing technology analysis research using SAO structures generally extracts SAO structures by analyzing the abstracts (Bergmann et al., 2008; Cascini et al., 2004b; Moehrle et al., 2005), but this paper uses the full text of patents. Although abstracts mainly describe important components of an invention, they cannot describe its purpose or provide technological details. The full text of patent documents includes the information regarding "problems to be solved" (Tseng, Lin, & Lin, 2007) and this information can represent technology details. Thus, to extract necessary SAO structures, we consider all parts of patent documents.

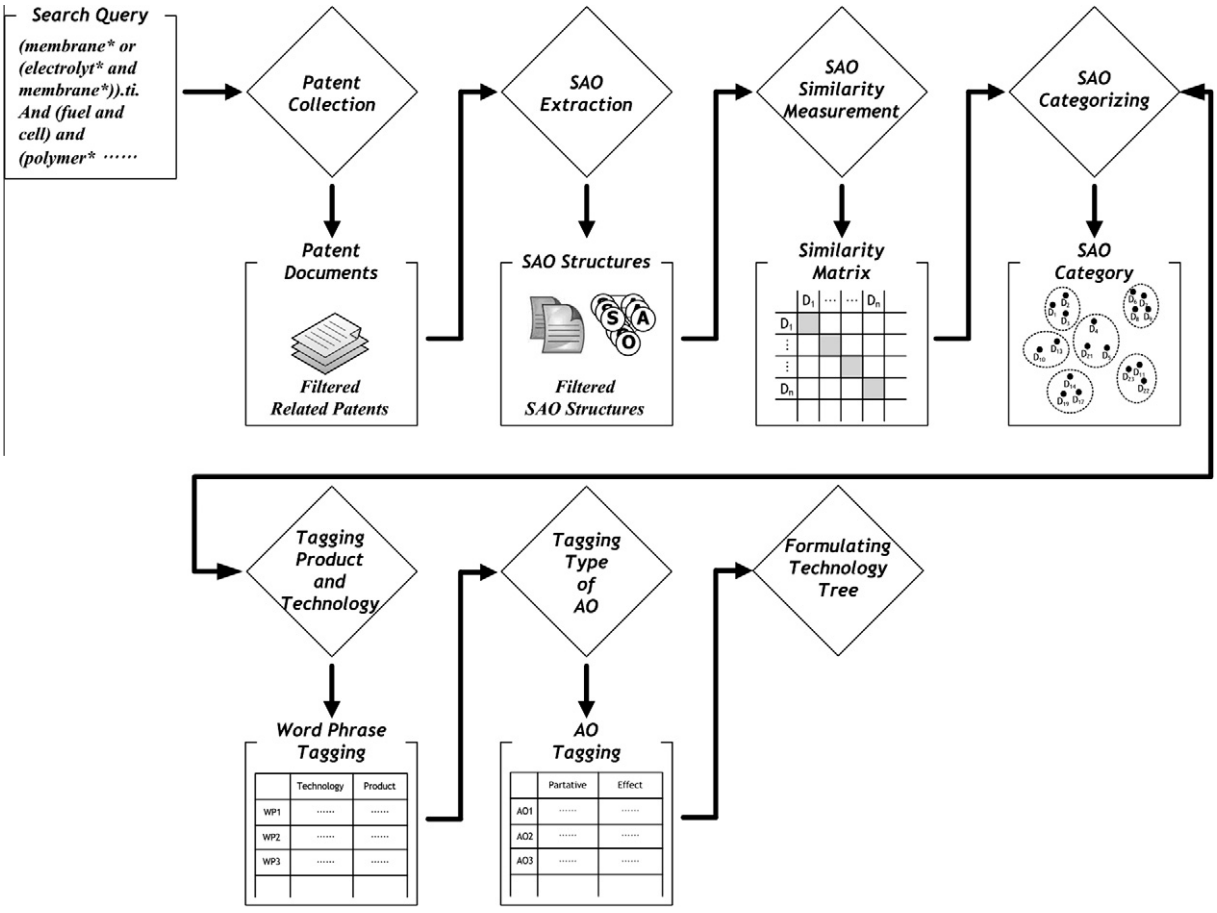


Fig. 1. A procedure for manipulating SAO structures.

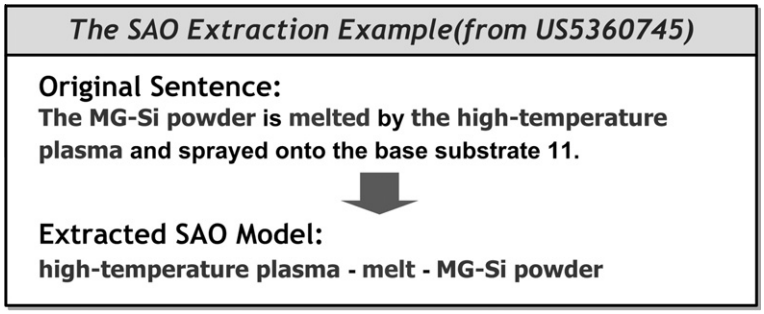


Fig. 2. An example of SAO extraction (Patent US5360745).

To automatically extract SAO structures using an NLP parser, we use the commercial linguistic analyzers Knowledgist2.5™ (www.invention-machine.com). By using NLP technology, we can extract SAO structures (Fig. 2).

3.1.3. Measuring SAO similarity

This step generates a similarity matrix of SAO structures. The purpose of measuring SAO similarities is to categorize similar word phrases and action-object (AO) combinations. In SAO structures, the subject and the object each contain a word phrase that represents a product, a technology, or a technological indicator. Because of synonyms, a word phrase can be represented in different ways. For example, 'PEFC' and 'PEMFC' both mean 'Proton Exchange Membrane Fuel Cell'. In addition, word phrases have generalization relationships among them. For example, the word phrases such as 'a-SiC:H', 'microcrystalline silicon', or 'Amorphous silicon' are specialized word phrases of 'silicon'. To allow analysis of a variety of word phrases, they should be clustered, tagged as a representative term, and managed by level of generalization. In the process of constructing a TechTree, the clustering step helps users obtain the various aspects of technology information by abstracting word phrases that contain technology information. For the same reason, AOs also should be clustered, tagged, and managed. For example, because 'manufacture gas diffusion electrode' and 'make gas diffusion layer' have the same meaning, they must be regarded as the same AO.

To help cluster word phrases and AOs, we use WordNet-based semantic sentence similarity measurement (Simpson & Dao, 2005) generate the similarity matrix of SAO structures. Using the C# library for the WordNet-based semantic sentence similarity measurement (Simpson & Dao, 2005), semantic similarity between two SAO structures can be computed. In most cases, the WordNet defines the concept hierarchies of most words, such as synonyms, hypernyms and hyponyms, by word meaning: for example, the chemical symbol for element 'Mg' is the same as 'magnesium' and the noun 'automobile' is a variant of the noun 'vehicle'. However, WordNet does not contain abbreviations that are strongly domain specific: for example, researchers use specialized acronyms, such as 'PEMFC' for 'Proton Exchange Membrane Fuel Cell' and 'CVD' for 'chemical vapor deposition'. Therefore, we modified the C# library source code to refer to a synonym set that can be grouped by investigating SAO structures of the data source. After calculating similarities, a similarity matrix is generated (Fig. 3).

3.1.4. Categorizing word phrases and AOs

After producing the similarity matrix, we use it to categorize word phrases and AOs. Several clustering algorithms can be used to categorize SAO structures; we adopted the *k*-means clustering algorithm, which is a method of cluster analysis which aims to partition *n* observations into *k* clusters in which each observation belongs to the cluster with the nearest mean (MacQueen, 1967). In *k*-means clustering, the best number *k* of clusters tends to show the smallest average distance between individual elements and cluster center, and the largest average distance average distance between cluster centers (Yoon & Park, 2004). However, a particular cluster may not be completely separable from others, and this means that the cluster is not different from others in an overall sense. The value *k* that gives the best clustering result is found by trial-and-error. Using a similarity matrix, we can cluster keyword phrases (Table 1).

	KP1	KP2	KP3	KP4	KP5	KP6	KP7	KP8	KP9	KP10	KP11	KP12	KP13	KP14	KP15	KP16	KP17	KP18	KP19	KP20	KP21	KP22	KP23	KP24	KP25	KP26
KP1	1.00	0.29	0.25	0.44	0.75	0.75	0.17	0.29	0.82	0.15	0.22	0.17	0.53	1.00	0.40	0.17	0.50	0.59	0.47	0.40	0.24	0.94	0.82	0.35	0.17	0.40
KP2	0.29	1.00	0.45	0.62	0.53	0.53	0.65	1.00	0.55	0.58	0.45	0.59	0.64	0.29	0.62	0.67	0.61	0.77	0.76	0.75	0.59	0.58	0.39	0.81	0.59	0.65
KP3	0.25	0.45	1.00	0.62	0.71	0.71	0.71	0.53	0.71	0.33	1.00	0.36	0.83	0.25	0.62	0.67	0.69	0.71	0.53	0.77	0.66	0.33	0.31	0.62	0.59	0.36
KP4	0.44	0.62	0.62	1.00	0.62	0.62	0.62	0.62	0.33	0.29	0.62	0.31	0.85	0.44	1.00	0.59	0.52	0.78	0.72	0.67	0.59	0.29	0.27	0.78	0.95	0.33
KP5	0.75	0.53	0.71	0.62	1.00	1.00	0.71	0.53	0.71	0.33	0.71	0.36	0.69	0.75	0.62	0.67	0.59	0.71	0.53	0.77	0.52	0.71	0.78	0.62	0.59	0.36
KP6	0.75	0.53	0.71	0.62	1.00	1.00	0.71	0.53	0.71	0.33	0.71	0.36	0.69	0.75	0.62	0.67	0.59	0.71	0.53	0.77	0.52	0.71	0.78	0.62	0.59	0.36
KP7	0.17	0.65	0.71	0.62	0.71	0.71	1.00	0.65	0.57	0.73	0.71	0.50	0.76	0.17	0.62	0.75	0.64	0.75	0.63	0.73	0.68	0.60	0.40	0.44	0.59	0.67
KP8	0.29	1.00	0.53	0.62	0.53	0.53	0.65	1.00	0.55	0.58	0.45	0.59	0.64	0.29	0.62	0.67	0.61	0.77	0.76	0.75	0.59	0.58	0.39	0.81	0.59	0.65
KP9	0.82	0.55	0.71	0.33	0.71	0.71	0.57	0.55	1.00	0.50	0.71	0.57	0.64	0.82	0.33	0.67	0.70	0.67	0.58	0.57	0.59	0.50	0.88	0.50	0.31	0.57
KP10	0.15	0.58	0.33	0.29	0.33	0.33	0.73	0.58	0.50	1.00	0.33	0.60	0.32	0.15	0.29	0.33	0.54	0.33	0.47	0.67	0.72	0.60	0.36	0.40	0.27	0.67
KP11	0.22	0.45	1.00	0.62	0.71	0.71	0.71	0.45	0.71	0.33	1.00	0.36	0.83	0.33	0.62	0.67	0.69	0.71	0.53	0.77	0.66	0.33	0.31	0.62	0.59	0.36
KP12	0.17	0.59	0.36	0.31	0.36	0.36	0.50	0.59	0.57	0.60	0.36	1.00	0.35	0.25	0.31	0.36	0.80	0.36	0.49	0.67	0.53	0.60	0.40	0.44	0.29	0.67
KP13	0.53	0.64	0.83	0.85	0.69	0.69	0.76	0.64	0.64	0.32	0.83	0.35	1.00	0.53	0.85	0.65	0.68	0.75	0.77	0.75	0.75	0.41	0.52	0.74	0.81	0.35
KP14	1.00	0.29	0.25	0.44	0.75	0.75	0.17	0.29	0.82	0.15	0.33	0.25	0.53	1.00	0.50	0.29	0.53	0.59	0.47	0.40	0.24	0.94	0.82	0.35	0.29	0.40
KP15	0.40	0.62	0.62	1.00	0.62	0.62	0.62	0.62	0.33	0.29	0.62	0.31	0.85	0.50	1.00	0.59	0.52	0.78	0.72	0.67	0.55	0.29	0.27	0.78	0.95	0.31
KP16	0.17	0.67	0.67	0.59	0.67	0.67	0.75	0.67	0.67	0.33	0.67	0.36	0.65	0.29	0.59	1.00	0.53	1.00	0.47	0.71	0.50	0.33	0.33	0.50	0.56	0.53
KP17	0.50	0.61	0.69	0.52	0.59	0.59	0.64	0.61	0.70	0.54	0.69	0.80	0.68	0.53	0.52	0.53	1.00	0.59	0.58	0.74	0.69	0.61	0.49	0.51	0.49	0.80
KP18	0.59	0.77	0.71	0.78	0.71	0.71	0.75	0.77	0.67	0.33	0.71	0.36	0.75	0.59	0.78	1.00	0.59	1.00	0.72	0.77	0.52	0.55	0.62	0.78	0.74	0.57
KP19	0.47	0.76	0.53	0.72	0.53	0.53	0.63	0.76	0.58	0.47	0.53	0.49	0.77	0.47	0.72	0.47	0.58	0.72	1.00	0.61	0.64	0.48	0.49	0.59	0.69	0.53
KP20	0.40	0.75	0.77	0.67	0.77	0.77	0.73	0.75	0.57	0.67	0.77	0.67	0.75	0.40	0.67	0.71	0.74	0.77	0.61	1.00	0.70	0.67	0.40	0.74	0.67	0.75
KP21	0.24	0.59	0.66	0.59	0.52	0.52	0.68	0.59	0.59	0.72	0.66	0.53	0.75	0.24	0.55	0.50	0.69	0.52	0.64	0.70	1.00	0.69	0.34	0.50	0.42	0.58
KP22	0.94	0.58	0.33	0.29	0.71	0.71	0.60	0.58	0.50	0.60	0.33	0.60	0.41	0.94	0.29	0.33	0.61	0.55	0.48	0.67	0.69	1.00	0.50	0.40	0.27	0.67
KP23	0.82	0.39	0.31	0.27	0.78	0.78	0.40	0.39	0.88	0.36	0.31	0.40	0.52	0.82	0.27	0.33	0.49	0.62	0.49	0.40	0.34	0.50	1.00	0.36	0.25	0.90
KP24	0.35	0.81	0.62	0.78	0.62	0.62	0.44	0.81	0.50	0.40	0.62	0.44	0.74	0.35	0.78	0.50	0.51	0.78	0.59	0.74	0.50	0.40	0.36	1.00	0.74	0.44
KP25	0.17	0.59	0.59	0.95	0.59	0.59	0.59	0.59	0.31	0.27	0.59	0.29	0.81	0.29	0.95	0.56	0.49	0.74	0.69	0.67	0.42	0.27	0.25	0.74	1.00	0.29
KP26	0.40	0.65	0.36	0.33	0.36	0.36	0.67	0.65	0.57	0.67	0.36	0.67	0.35	0.40	0.31	0.53	0.80	0.57	0.53	0.75	0.58	0.67	0.90	0.44	0.29	1.00

Fig. 3. An example of similarity matrix for keyword phrases (KP).

Table 1
An example of clustered word phrases.

Cluster number	Raw word phrase
49	Anode Anode and cathode Cathode and anode Anode compartment Cathode
19	Carbon Carbon fiber woven cloth Carbon fiber based sheet Carbon black Carbon blacks Mixture of carbon-containing compound Skeleton section of carbon-containing compound Carbon-containing compound Carbon-containing phase

Table 2
Types of keyword phrase.

Type	Definition	Examples
Product	Products or product components are all items needed for technology activities that can be procured in the market, including services (Guglielmi et al., 2010)	Amorphous silicon thin film, PEMFC
Technology	A technology is defined as the technical know-how that is required for the design, manufacture and test of a space product, including also all related processes, manufacturing process or method (Guglielmi et al., 2010)	CVD, MOCVD
Material	Materials which are not only belong to specific technologies or products, used in various area (Hirtz, Stone, McAdams, Szykman, & Wood, 2002). e.g. gas, liquid, solid, plasma, mixture	Water, gas, air, powder, oxygen, hydrogen
Technology attribute	The measurement of purpose of developing technology, which is needed to increase or decrease	Conductivity, thickness of film

3.1.5. Identifying types of word phrases

After clustering SAO structures, we identify the types of clustered word phrases based on their characteristics. Word phrases are divided into four types: product, technology, material, and technology attribute (Table 2).

Table 3
An example of action reference table for identifying the type of a word phrase.

Subject	Action	Object
Technology	Manufacture produce form supply	Product
Product	Include have composed of comprise supply	Product
Technology product	Use convey supply	Material type
Technology product material type	Intensify increase measure stabilize	Technology attribute

To identify the types of keyword phrase in a SAO structure, we use an SAO linked list that represents relationships among SAO structures (Fig. 4). Generally, a product is manufactured using a technology, and a technology attribute is affected by a product or a technology. In addition, materials are used and owned by a product or a technology. For example, CVD is a technology for manufacturing amorphous silicon thin film and amorphous silicon thin film affects conductivity, which is a technology attribute.

To reliably identify the type of a word phrase, we consider the action word that links the subject to the object. Based on the type of action, we can identify the type of word phrase used in the subject and the object. To this end, we need a reference table that defines the types of action (Table 3). In an SAO structure, if the subject and the object are word phrases regarding a product type, they are likely to be linked by an action word such as 'include', 'comprise', or 'compose'. If the subject is a product or technology type, and the action word is 'intensify' or 'increase' then the object is highly likely to be a type of technology attribute. By consulting Table 3, the developer of a TechTree can manually identify the type of word phrase by using the most generalized terms.

3.1.6. Identifying the type of AO

In this step, we identify the types of categorized AOs. AOs are classified into three types: Partative, Effect, and Attribute (Table 4). The Partative type of AO represents an inclusion relationship between products or technologies. This type contains an action word such as 'contains' that represents an inclusion relationship (Cascini et al., 2001) and an object word that refers to a product or a technology. The Effect type of AO represents how the subject affects the object. This type is constructed as the combination of an action word such as 'absorb' and any type of key phrase (KP).

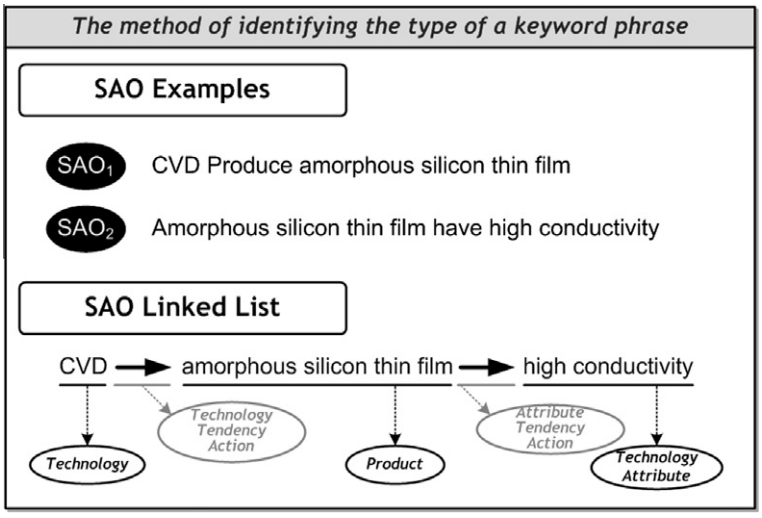


Fig. 4. An Example of SAO Linked List for identifying the type of a word phrase.

Table 4
AO types.

The type of AO	Definition	Action	KP type of object
Partative type	AO representing inclusion relationship between products or technologies	Include, have, composed of, comprise, supply, assembles	Product technology
Effect type	AO representing how the subject word affects the object word	Absorb, Accumulate, Cleans, Condense, Cools, Destroys, Detects, Erodes, Evaporate, Extracts, FreezesBoil, Heats, Holds, Mixes, Moves, Polishes, Produces, Removes, Separates	Any type of KP
Attribute type	AO representing how the subject word changes the technology attribute	Stabilizes, Vibrates, Intensify, Increase, Measure, Stabilize	Technology attribute

The Attribute type of AO represents how the subject word changes the technology attribute. This type contains an action word such as 'increase' and a technology attribute. The process identifying AO types depends on the type of combination between the action word and the type of the most generalized word phrase used in an object word. Based on the combinations, the type of an AO is identified and manually modified as needed.

3.2. A method for construing a technology tree using the SAO structures

3.2.1. Three dimensions of technology tree

To construct the proposed TechTree, this paper considers three dimensions. The first dimension is the generalized level of SAO structures. Within this dimension, clustered word phrases and AOs are generalized to three levels. First level includes raw terms; terms of this level are from raw SAO structures which are not tagged as representative terms, and which represent detailed technology information. The second level is a representative term, which generalizes raw terms, and which represents main technology information. The third level includes the most generalized terms, which incorporate representative terms and represent the overview of technology information. For example, terms such as 'DMFC', 'hydrogen–oxygen fuel cell' and 'PEMFC' can be expressed in various forms and generalized as 'Fuel Cell' (Table 5).

The second dimension of the TechTree is the technology meta-information. Because SAO structures are extracted from patent documents, SAO structures can be linked to the bibliographic information of patent documents, including publication date, assignee, and assignee nationality. By using information of publication date, we can construct a TechTree that includes technologies developed in a specific period. The information of assignee or assignee nationality can be used to develop a TechTree that represents technology developed by specific assignees or countries, individually. In addition, a TechTree can be developed using all information or just necessary information.

Table 5
Example of generalized level of key phrase.

Raw term	Representative term	Generalized term
Direct methanol fuel cells	DMFCs	Fuel Cell
Direct methanol PEM fuel cell		
Direct-methanol fuel cell		
Hydrogen–oxygen fuel cell	Hydrogen–oxygen fuel cell	
Hydrogen/air fuel cell		
Hydrogen/oxygen fuel cells		
Hydrophilic fuel cell electrode		
Polymer electrolyte fuel cell	PEMFC	
Polymer electrolyte membrane fuel cell		
PEM fuel cells		

The third dimension of the TechTree is a perspective of technology Information. The perspectives considered are a product view, a technology view, and a function of technology. Based on these perspectives, three types a diagram are possible. The diagram of a product view represents the relationship between products and product components. The diagram of a technology view represents the relationship among technologies used for a product or a product component. The diagram of a function of technology represents the functions which are provided by a product or a technology, and the relationships among functions.

3.2.2. The types of a technology tree diagram

Based on the perspectives of technology information, we can formulate the diagrams divided into three types: Product-centered, Technology-centered, and Function-centered. In addition, depending on the application of a TechTree, the diagrams provide technology information that considers technology meta-information.

To construct a Product-centered TechTree diagram (Fig. 5(a)), the SAO structures of a partative type are used. The product-centered word phrases are used as node of the TechTree. In the diagram, 'Fuel Cell' is the generalized word phrase of 'PAFC', 'PEMFC' or 'DMFC' and 'Fuel Cell' is linked with the word phrase regarding product components such as 'MEA', 'membrane', 'electrocatalyst', and 'gas diffusion electrodes'. These linking relationships are noted in SAO structures (Fig. 5(a)).

A Technology-centered TechTree (Fig. 5(b)) expresses the relationships among the technologies used for products and product components; it is based on the relationships among products and product components. To construct a Technology-centered TechTree diagram, two types of SAO structures are analyzed. The first type is the SAO structure that contains a product type word phrase in the subject, a partative type in the action, and a technology type word phrase in the object. In this SAO structure, a technology type word phrase in the object is represented as a node of a TechTree. In a Technology-centered TechTree, because the diagram is based on the relationships used in a product based diagram, the technology type word replaces a product type word phrase which is a node of a Product-centered diagram. The product type word phrase is the subject of an SAO structure that contains a technology type word phrase in its object. The second type is an SAO structure that contains a technology-related word phrase such as 'method', 'invention' and 'process' in the subject and has an effect-centered AO that describes what the subject provides. In the SAO structure, the AO is considered to be a node of a TechTree. In Fig. 5(b), 'cooling fuel cell' is a technology for 'fuel cell' product and 'MEA' has related technologies such as 'sputter-depositing catalyst Technology' and 'eliminate carbon'. Because 'MEA' is a product component of 'fuel cell', the two technologies related to 'MEA' are connected to 'cooling fuel cell' technology.

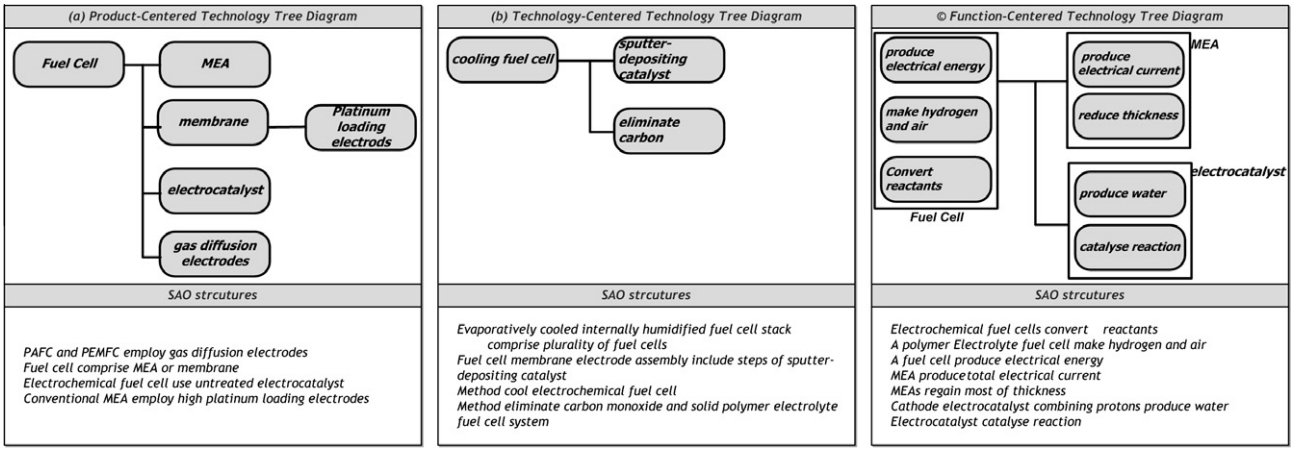


Fig. 5. An example of TechTree diagrams: (a) product, (b) technology, and (c) function of technology.

A Function-centered TechTree (Fig. 5(c)) expresses the relationships between functions which are provided by products or technologies. This diagram is also based on the relationships between products and product components. Generally, superordinate product components provide the functions that represent the purposes of a product, and subordinate product components provide the functions that represent the technological effect that accomplishes the purposes. In a Function-centered TechTree diagram, if the word phrase, tagged as a product or technology type, is used as a subject in an SAO structure, the AO is used as a node of the TechTree. In Fig. 5(c), the product 'fuel cell' has functions 'produce electrical energy', 'make hydrogen and air', and 'convert reactants'. These functions are the purposes of 'fuel cell'. 'MEA', which is a product component of 'fuel cell', has functions 'produce electrical current' and 'regain most of thicknesses'. In addition, 'electrocatalyst' has functions 'produce water' and 'catalyze reaction'. These functions express the technological effect that supports the functions of 'fuel cell'.

4. Case study: Proton exchange membrane fuel cells

To illustrate the proposed approach, we use it to develop a TechTree regarding PEMFC technology. PEMFCs constitute a next-generation energy technology; they are being developed for applications in transport, and to stationary and portable cells. When hydrogen reacts electrochemically with oxygen, chemical energy is generated; PEMFCs transform this chemical energy to electrical energy. A PEMFC consists of a stack of basic units (cells) for generating electricity and a cell stack consists of six parts (Fig. 6).

Qualitative patent analysis of target technology has already been performed by the Korean Intellectual Property Office (KIPO., 2006). We choose the patent set from the report, compare the result obtained by applying the proposed approach to those obtained by KIPO, and test whether new technological findings can be added by a TechTree of the relevant patents.

4.1. Data collection

As a first step, US patents related to PEFMC technology were collected from the WIPS database (www.wips.co.kr). The collected patents were published from January 1, 1985 to May 31, 2005. The patent set was already selected by KIPO; we removed irrelevant patents to optimize the time of analysis. One hundred and ten patents were used to construct the TechTree of PEMFC (Table 6).

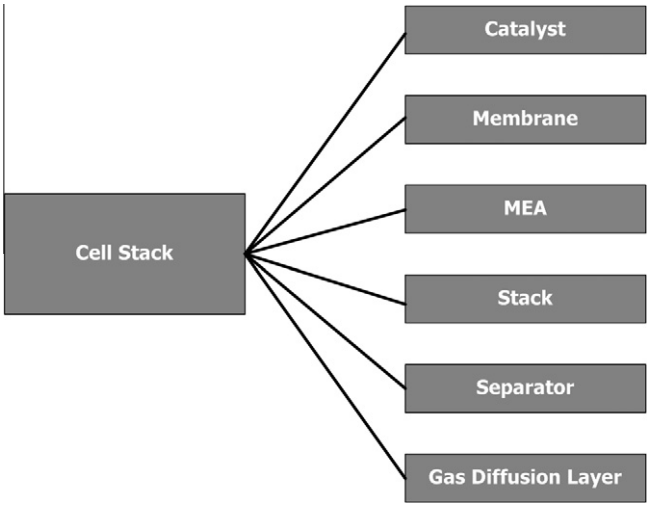


Fig. 6. The taxonomy of PEMFC technology (KIPO, 2006).

4.2. SAO processing

After collecting patent set, we processed SAO structures to utilize the sources of a TechTree. First, we extracted SAO structures after syntactic analysis of patent text. We used Knowledgist™ 2.5 to extract them from the full text of patent documents.

As the result of the syntactic analysis, we obtained 14289 SAO structures; 1592 remained after eliminating duplicated and irrelevant ones. To categorize keyword phrases and AO combinations, we used WordNet-based sentence similarity (Simpson & Dao, 2005) to construct the similarity matrix (Fig. 7) and used using *k*-means clustering to categorized. The available SAO structures included 1481 KPs; 702 remained after categorizing them. Among the remaining KPs, 261 were product type, 59 were technology type, 101 were attribute type, and 245 were material type. In the case of AO combinations, 620 AOs were partative type, 679 were effect type, and 293 were attribute type.

4.3. Analysis of the technology tree

After processing SAO structures, we construct the TechTree regarding PEMFCs. To this end, we developed a prototype system, called TechTree Analyzer (Fig. 8), to construct a TechTree based on the proposed approach. The prototype is implemented with Java

Table 6
Patent retrieval query and patent set.

Part of cell stack	Patent retrieval query	Patent set
Catalyst	(electrode* or cathode* or anode* or catalyst*).ti. and (fuel and cell*) and (polymer* or proton*) and @AD>=19850101<=20060531)	7045479, 6991870, 6899970, 6890670, 6855453, 6855452, 6815114, 6773429, 6670298, 6576208, 6528201, 6479182, 6479181, 6171721, 6057056, 5869202, 5795672, 5766788, 5716437, 5407756
Membrane	(membrane* or (electrolyte* and membrane*)).ti. and (fuel and cell) and (polymer* or proton*) and @AD>=19850101<=20060531)	5449697, 6037514, 5679482, 5817718, 5861222, 6444343, 5919583, 5981097, 6562446, 6716548, 6451470, 6509441, 6482763, 7049020, 6864006, 6893763, 6902839, 6946362, 7014931, 6878475
MEA	(MEA or (membrane* and electrode*)).ti. and (fuel and cell*) and (polymer* or proton*) and @AD>=19850101<=20060531)	5372896, 5523177, 5702837, 6110333, RE37307, 6020083, 6605391, 6306536, 6514561, 6523699, 6790553, 6756147, 6572790, 6670065, 6425944, 6774150, 6706435, 7005397, 6875537, 7001446
Separator	(separat* or bipolar* or (flow and field)).ti. and (fuel and cell*) and (polymer* or proton*) and @AD>=19850101<=20060531)	5277996, 5599638, 5500292, 6074692, 6183668, 6794078, 6475656, 6783883, 6472094, 6251308, 6425993, 7052792, 6706437, 6733914, 6713207, 6852438, 6827747, 6699593, 6964823, 6847518
Stack	stack*.ti. and (fuel and cell*) and (polymer* or proton*) and @AD>=19850101<=20060531)	5234777, 5501915, 5595834, 6050331, 5702839, 6080503, 6153327, 6153323, RE37433, 6558824, 6638650, 6432571, 6500577, 6770394, 6376129, 7014939, 6805989, 7001686, 6926995, 7052796
Gas diffusion layer	(gas and (diffus* or lay*)).ti. and (fuel and cell*) and (polymer* or proton*) and @AD>=19850101<=20060531)	5783325, 5863395, 6127059, 6566004, 6716549, 6949308, 6803143, 6733915, 7056613

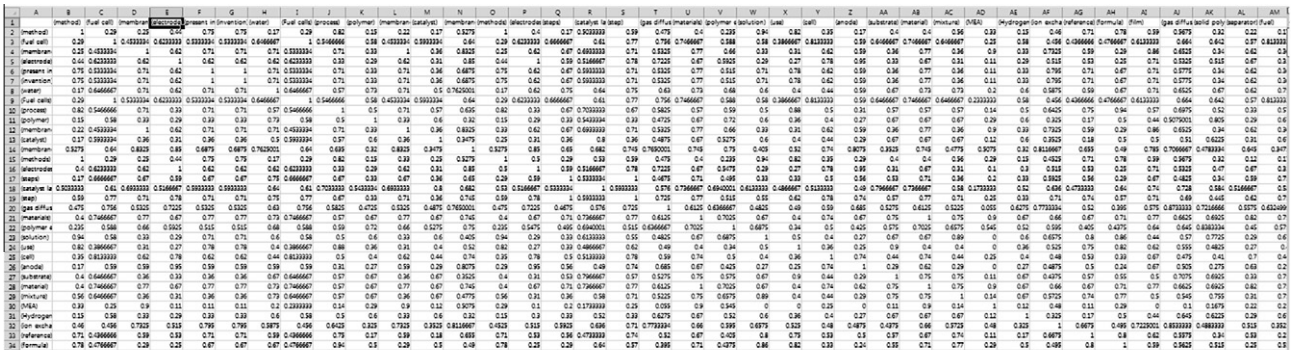


Fig. 7. Similarity matrix of keyword phrase.

and .NET framework. We used JGraph 5 API to visualize TechTree models, and MySQL database to process patent information with SAO structures. To measure the similarity among the SAO strucutures goal descriptions, we deployed the WordNet-based semantic similarity module (Simpson & Dao 2005) as a .NET web service. We used the web service by using the Java Axis Service client module. As a development platform environment, we used Eclipse for Java IDE and Visual Studio 2008 for .NET framework. We used Java 1.6.0 as the Java virtual machine.

The prototype system provides an interface to users to allow them to select technology perspectives, levels of generalization, and technology meta-information. The system accesses the database that stores patent information with SAO structures, and uses it to draw a TechTree diagram based on the request of users.

The TechTree Analyzer starts with a basic diagram when user requests that it generate a TechTree. The basic diagram represents product, technology and function relationships (Fig. 9). The product relationships demonstrate that 'Catalyst layer' has sub-components such as 'ion exchange polymer', 'electrode catalyst', and 'electrolyte catalyst'. The technology relationships represent how technologies are related to the product components. We can grasp that 'catalyst layer' is related to technologies regarding 'how to use air brush' and 'how to stimulate absorption of hydrogen and oxygen molecules'. Finally, the function relationships show why product components were developed. In the case of 'catalyst layer', 'to have high porosity or stable cavities' was an important reason for developing it. In fact, optimizing the size of cavities in the catalyst layer increases the performance and mechanical stability of the catalyst layer.

4.3.1. Product-centered technology tree diagram

A Product-centered TechTree diagram represents the structural relations among product components. This diagram is generated using partative type SAO structures. A cell stack of PEMFC divided into two main product components: a membrane electrode assembly (MEA) and a separator, and each of these components includes sub-components. The prototype system analyzed SAO structures and generated a generalized product-based TechTree diagram that includes this cell-stack structure (Fig. 10(a)). This generalized diagram includes a more-detailed diagram called a Representative Product-based TechTree diagram that tags keyword phrases (Fig. 10(b)). This diagram provides more-detailed information about structural relations than does the generalized diagram. The separator (Fig. 10(c)) includes some subcomponents such as 'channel' or 'flow channel'. In fact, a separator is a component that separates MEAs in the PEMFC, and its path is a key component to transmit fuel gas and oxygen between MEAs. This path is required for the PEMFC

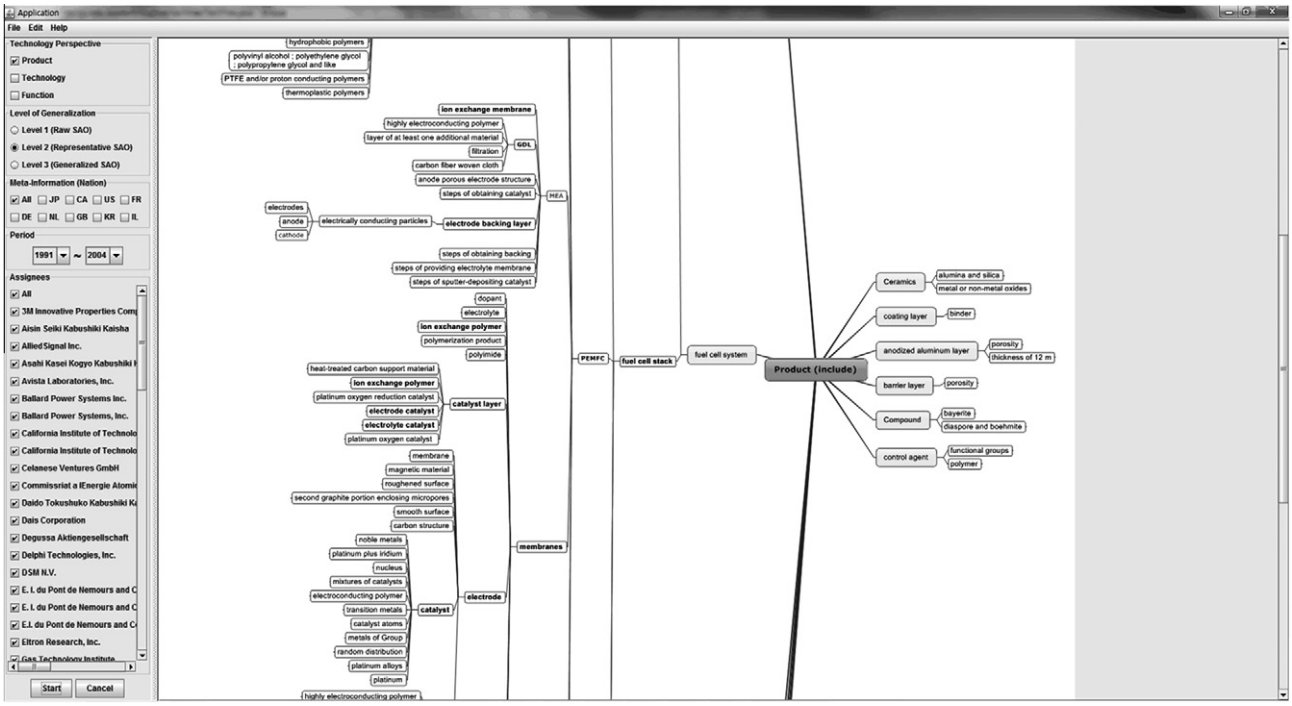


Fig. 8. Screenshot of TechTree Analyzer.

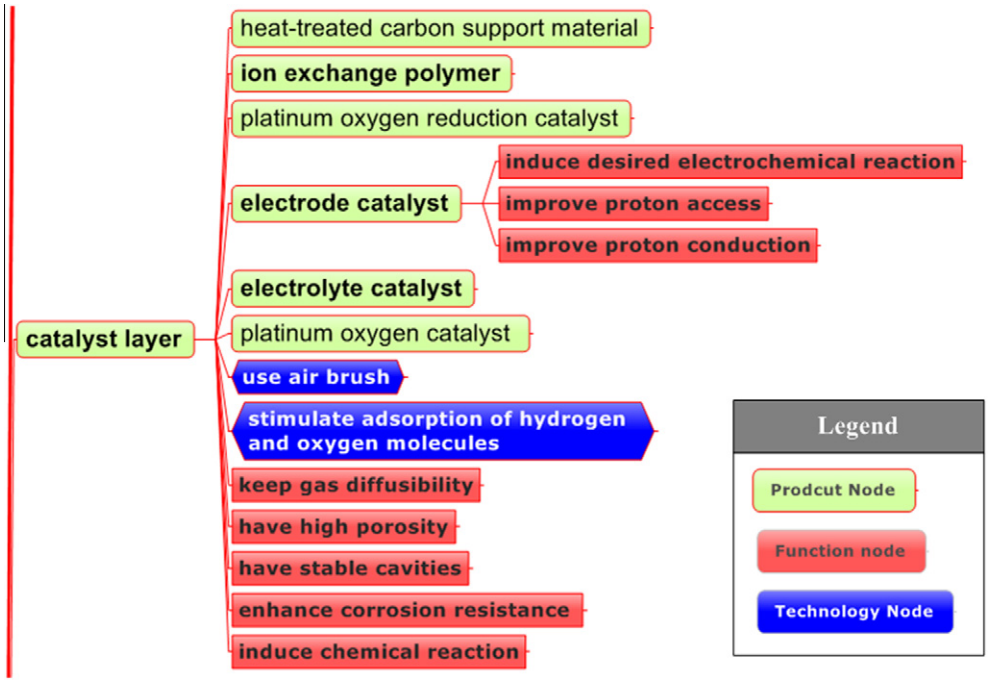


Fig. 9. An example of basic diagram regarding catalyst layer. Green nodes: product relationships; blue nodes: technology relationships; red nodes: function relationships. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

to achieve high thermal and electricity conductivity with low gas-permeability. The performance of a separator depends on how the channel is developed.

The product-centered TechTree diagram can be also used for understanding the key components of the membrane. The diagram (Fig. 11) represents two types of membrane: perfluoro-based polymer and carbon-based polymer. Membranes represented in the upper boxes are types of perfluoro-based polymer membrane, such as 'fluorinated compound', 'sulfonated polyimide membrane', and 'Nafion'. Membranes in the lower boxes are types of carbon-based polymer membrane, including 'carbon compound', 'perfluorocarbon compound', and 'hydrocarbon membrane'.

4.3.2. Technology-centered technology tree diagram

A Technology-centered TechTree diagram represents technology information related to product components. This diagram is

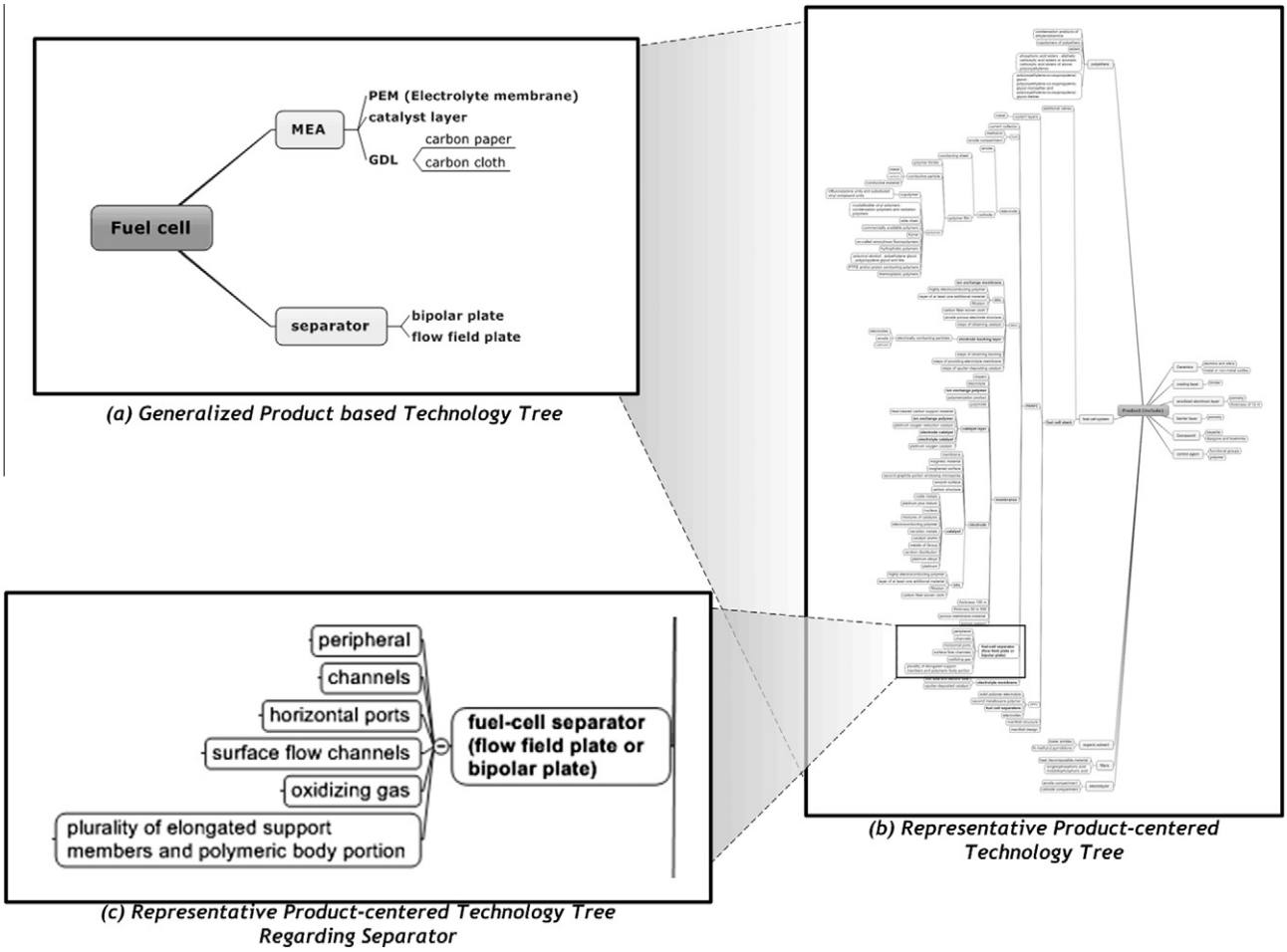


Fig. 10. Product-centered TechTree diagram.

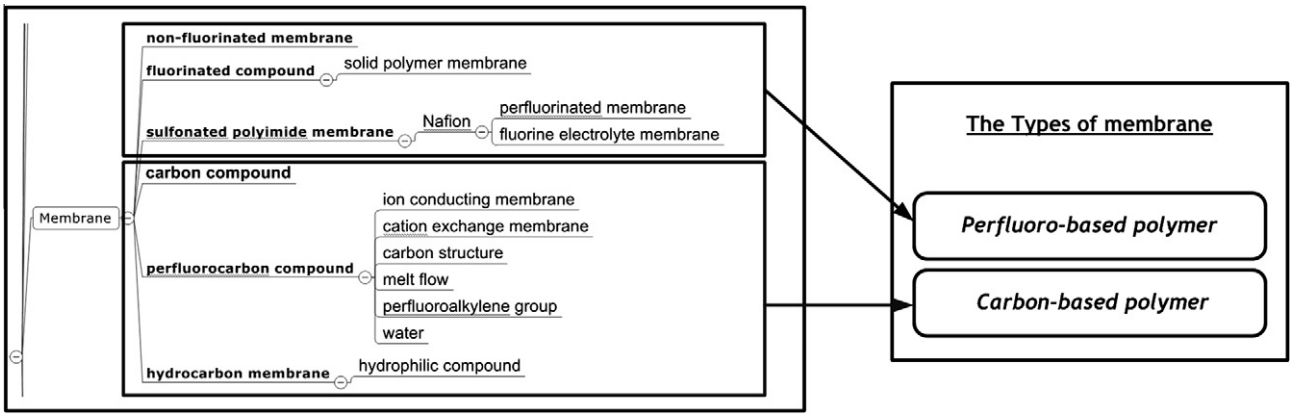


Fig. 11. Types of membrane in a PEFMC.

generated based on technology-centered keyword phrases and effect-centered type SAO structures. Because effect-centered AOs are generally related to a technology principle, we can infer the technologies used in a specific area by categorizing effect-centered AOs. In our case, the product component 'catalyst' has seven technology-centered nodes (Fig. 12). The nodes can be categorized into three technology types related to catalysts.

The first type is a technology regarding improvement of electricity performance; the four nodes regarding 'to promote reactions' are of this type. A high reaction rate of catalysts in PEMFC increases its power output. This type of technology is being actively investigated.

The second technology type includes methods of overcoming the CO poisoning problem, which occurs when CO produced by a chemical reaction in PEMFC adsorbs to the platinum and degrades its catalytic ability. CO poisoning reduces the reaction rate of poisoned catalysts decreases the durability of PEMFCs. To solve this problem, the research regarding platinum's resistance to CO is

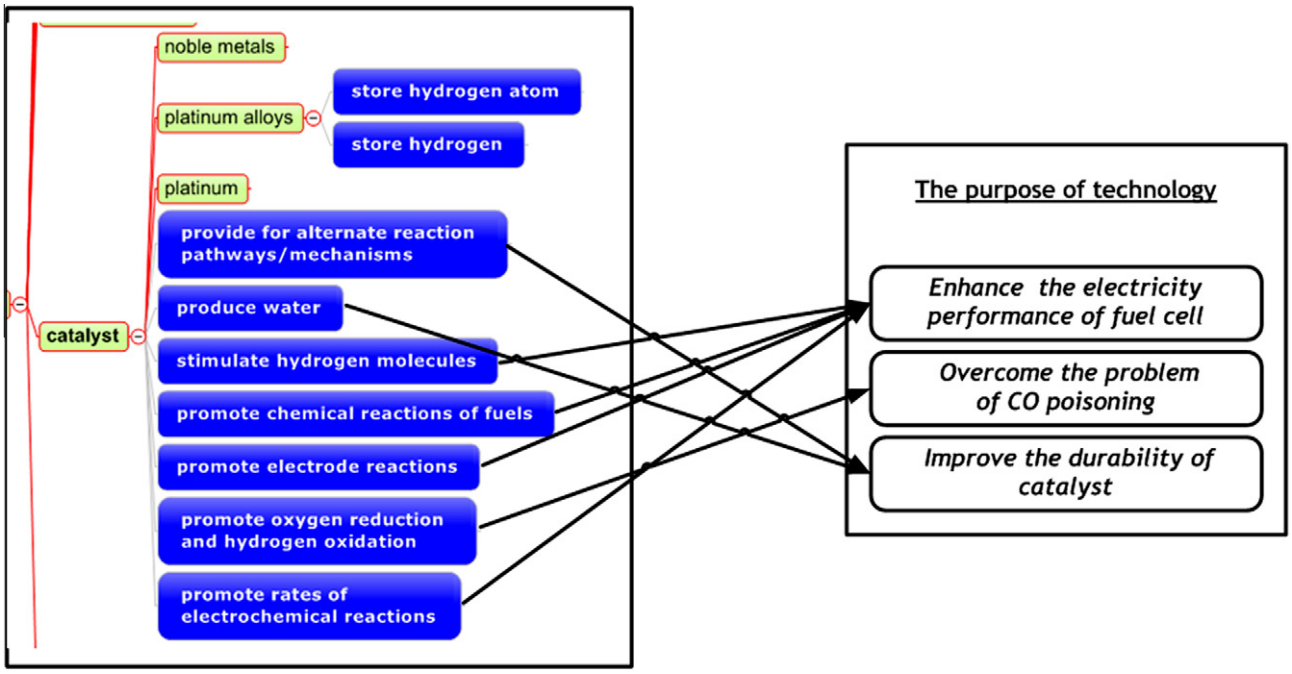


Fig. 12. Technology-centered TechTree diagram of "catalyst".

being actively conducted and 'to promote oxygen reduction and hydrogen oxidation' is one option for solving the CO poisoning problem.

The third technology type concerns the durability of catalysts. Methods for eliminating water produced in chemical reactions re important for improving the durability of catalysts. 'To provide alternative pathways for reaction' is one method to eliminate water.

4.3.3. Function-centered technology tree diagram

A Function-centered TechTree diagram presents the technological purposes of product components. This diagram is generated based on attribute-centered SAO structures. Because these type combinations are connected to the technological purpose, we can transform them into technological indicators. In our case, the product component 'membranes' has 13 function-centered nodes (Fig. 13). From the nodes, we extracted five technological indications: 'Ion conductivity', 'Unit cost', 'Degree of methanol cross-over', 'Mechanical strength', and 'Decrease of resistance'. In fact, improvement of ion conductivity is regarded as a very important technology development theme. In the case of 'cost', decreasing the cost of the membrane is an important goal of technology development to achieve deployment of PEMFCs on a commercial scale. 'Methanol cross-over' degrades the efficiency of PEMFCs, so preventing the phenomenon is very important. 'Mechanical strength' and 'decrease of resistance' are related to membrane thickness, which has important effects on its strength and resistance; reducing thickness increases the membrane's conductivity and power efficiency, but decreases its mechanical strength and resistance. Therefore optimizing membrane thickness is very important.

4.3.4. Dynamic analysis of a technology tree using technology meta-information

Finally, we also propose dynamic analysis of a technology tree using technology meta-information. For dynamic analysis, we use two technology meta-information: assignee and filing date.

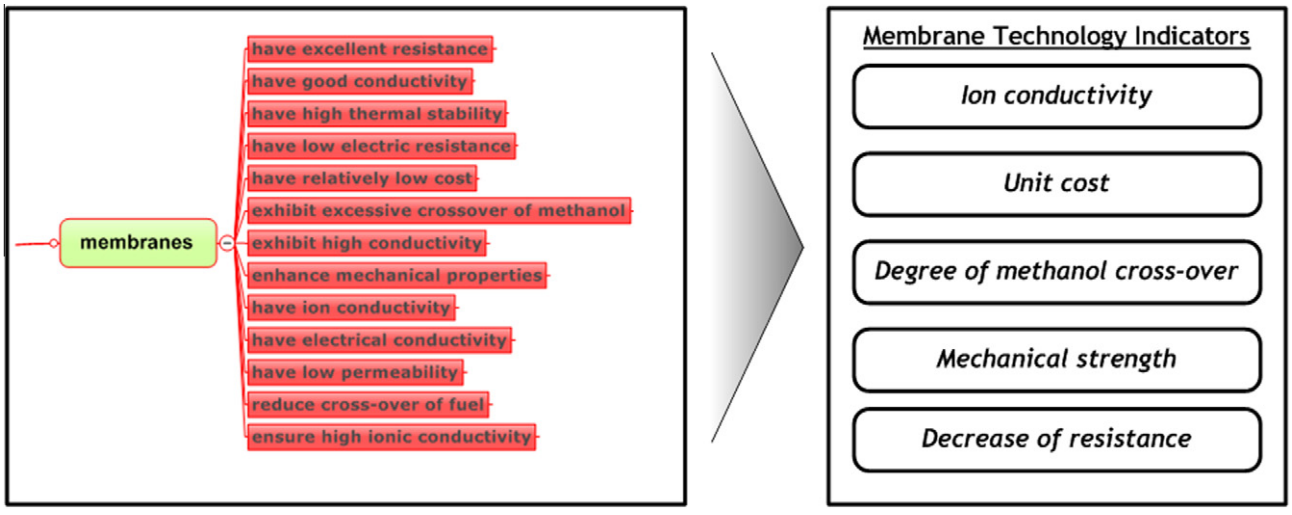


Fig. 13. Function-centered technology tree diagram of "membrane".

First, we compare the characteristics of membrane of two companies: Du Pont and Gore. Du Pont and Gore are representative companies for perfluoro-based and carbon-based membrane, individually. Because of the differences between two types of membranes, the functions of membranes show differences (Fig. 14). In case of Du Pont, 5 functions are considered for developing membrane: conductivity related, physical durability related, crossover related, heat stability related, and porous structure related functions. Especially, we can find that Du Pont make efforts to reduce methanol crossover problem. Whereas, Gore more focus on physical durability and porous structure. Almost 60% of functions of Gore's membrane are related to increase mechanical strength.

Second, we use time period information to compare the membranes produces before 2001 to those produced after 2001 (Fig. 15). Before 2001, membrane research focused on developing conductivity and physical durability. After 2001, these two functions became less important and new functions such as heat stability, appeared. This comparison shows that the research

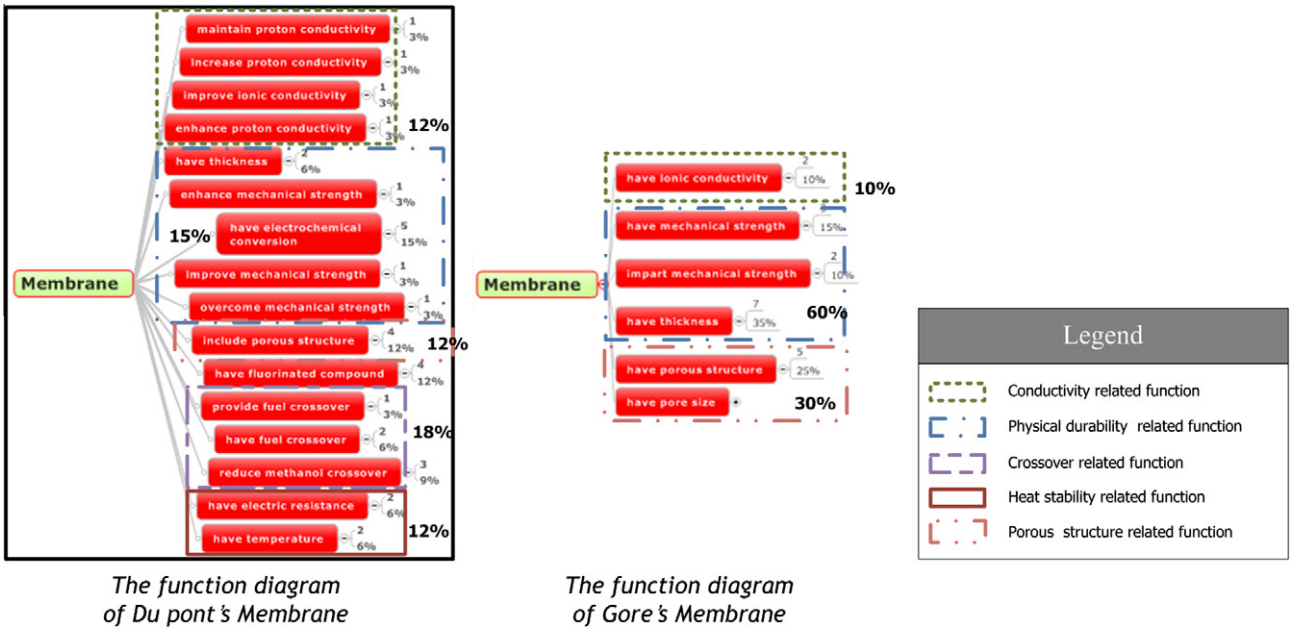


Fig. 14. The function diagrams of membranes for Du Pont and Gore.

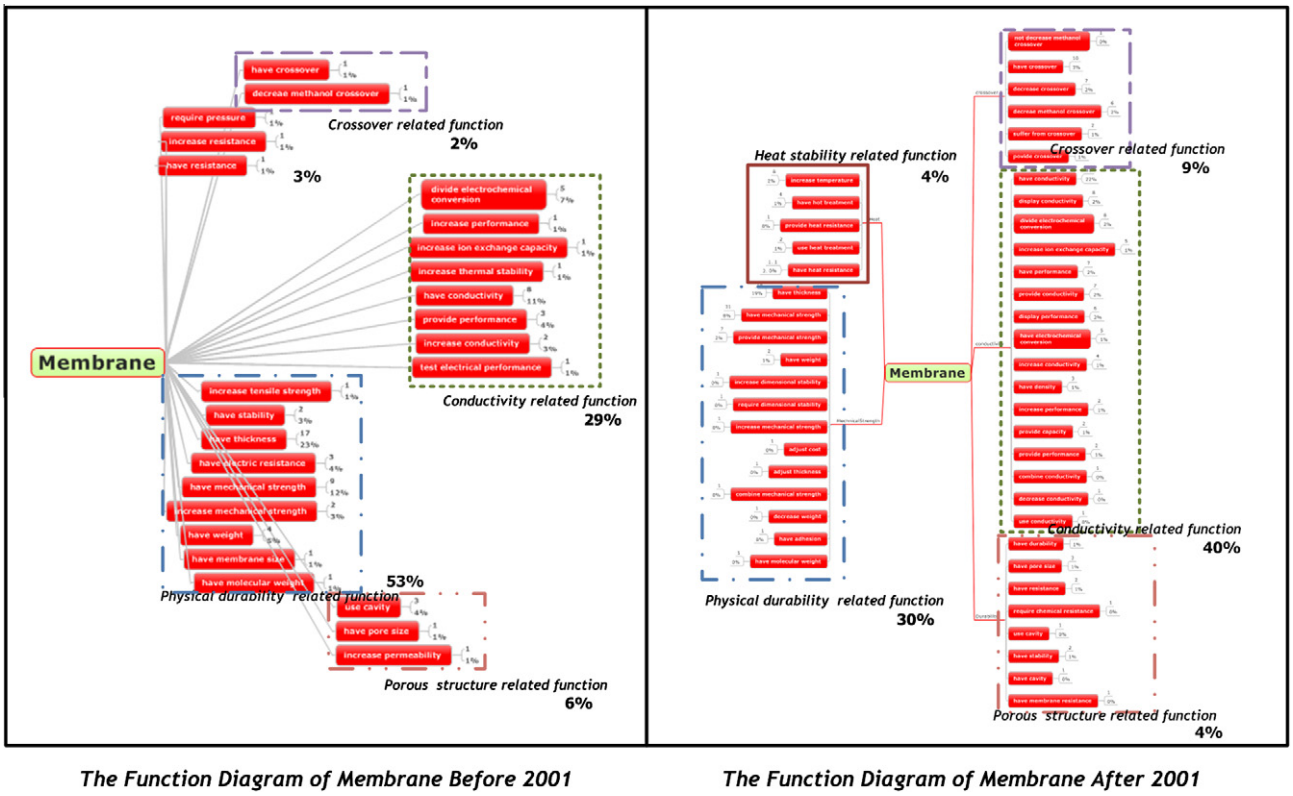


Fig. 15. The function diagram of membranes for two periods.

themes of membranes are diversifying, and that conductivity and physical durability may have been optimized.

5. Conclusion

In spite of the importance of TechTrees as tools for technology planning, little research has been conducted on their development. This shortage has caused the limitation that dynamically updating and maintaining TechTrees is difficult; this is a serious problem because technology is advancing so quickly. To remedy these problems, we presented an SAO-based text mining approach to developing TechTrees. The proposed approach extracts SAO structures from patent documents, and analyzes them to develop a TechTree. Text-mining techniques are used to extract SAO structures, which are then categorized by similarity, and are identified by the type of technological implications that they represent. To demonstrate the feasibility of the proposed approach, we developed a case study for PEMFC technology.

The proposed approach provides a quantitative method that uses patent information, rather than experts' knowledge, and that considers technology meta-information when constructing the TechTree. This approach will be used as a useful tool to help researchers or R&D policy makers to conduct technology planning. Furthermore, we expect that it can be also used for strategic R&D planning by exploiting various types of technology meta-information. For example, in the merger and acquisition process the decision maker can utilize the method to identify technological advantages by comparing the TechTrees of different companies.

To make the proposed approach more practical, three challenges must be addressed. First, a more efficient method extracting SAO structures should be developed. Knowledgist™ is almost the only tool available for extracting SAO structures. Although it is easy to use, it extracts many irrelevant SAO structures. This problem means that time must be spent removing irrelevant SAO structures and reduces the reliability of the TechTree constructed using the proposed approach. A better SAO extractor can guarantee a TechTree of better quality. Second, well-defined technology thesauri or ontologies should be constructed. In the proposed approach, we manually analyzed technological word phrases to identify their types. Well-defined thesauri would be more adequate for resolving identification of technology words. Finally, to support the development of a TechTree systematically, an analysis system should be constructed. Currently, we use Knowledgist™ to extract SAO structures and to develop a system of constructing a TechTree. However, the current system cannot fully support the overall process of constructing a TechTree. If an optimized system that includes a technological database can be developed, it would improve the practicality of the proposed approach.

Acknowledgements

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